

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                           |                        |
|---------------------------|------------------------|
| Product Description:      | <b>4-Fluorostyrene</b> |
| Cat No. :                 | <b>L08252</b>          |
| CAS No                    | 405-99-2               |
| Molecular Formula         | C8 H7 F                |
| REACH registration number | -                      |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                      |                          |
|----------------------|--------------------------|
| Recommended Use      | Laboratory chemicals.    |
| Uses advised against | No Information available |

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

**E-mail address** [begel.sdsdesk@thermofisher.com](mailto:begel.sdsdesk@thermofisher.com)

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

**CLP Classification - Regulation (EC) No 1272/2008**

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## Physical hazards

Flammable liquids

Category 3 (H226)

## Health hazards

Skin Corrosion/Irritation

Category 2 (H315)

Serious Eye Damage/Eye Irritation

Category 2 (H319)

Specific target organ toxicity - (single exposure)

Category 3 (H335)

## Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Warning

## Hazard Statements

H226 - Flammable liquid and vapor

H319 - Causes serious eye irritation

H315 - Causes skin irritation

H335 - May cause respiratory irritation

## Precautionary Statements

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

## 2.3. Other hazards

Contains a known or suspected endocrine disruptor

Contains a substance on the National Authorities Endocrine Disruptor Lists

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component       | CAS No   | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008               |
|-----------------|----------|-------------------|----------|-----------------------------------------------------------------|
| 4-Fluorostyrene | 405-99-2 | EEC No. 206-975-0 | 97       | STOT SE 3 (H335)<br>Skin Irrit. 2 (H315)<br>Eye Irrit. 2 (H319) |

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|                                         |         |           |     |                                                                                                                                                                      |
|-----------------------------------------|---------|-----------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1,2-Benzenediol, 4-(1,1-dimethylethyl)- | 98-29-3 | 202-653-9 | 0.1 | Acute Tox. 4 (H302)<br>Acute Tox. 4 (H312)<br>Skin Corr. 1B (H314)<br>Skin Sens. 1 (H317)<br>Eye Dam. 1 (H318)<br>Aquatic Acute 1 (H400)<br>Aquatic Chronic 2 (H411) |
|-----------------------------------------|---------|-----------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Component                               | Specific concentration limits (SCL's) | M-Factor | Component notes |
|-----------------------------------------|---------------------------------------|----------|-----------------|
| 1,2-Benzenediol, 4-(1,1-dimethylethyl)- | -                                     | 1        | -               |

|                           |   |
|---------------------------|---|
| REACH registration number | - |
|---------------------------|---|

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.                                                        |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Get medical attention.                                                                                                   |
| <b>Inhalation</b>                         | Remove from exposure, lie down. Remove to fresh air. If not breathing, give artificial respiration. Get medical attention.                       |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### 4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

### 4.3. Indication of any immediate medical attention and special treatment needed

|                    |                                                 |
|--------------------|-------------------------------------------------|
| Notes to Physician | Treat symptomatically. Symptoms may be delayed. |
|--------------------|-------------------------------------------------|

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Water mist may be used to cool closed containers. Chemical foam. Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

No information available.

### 5.2. Special hazards arising from the substance or mixture

Flammable. Vapors may travel to source of ignition and flash back. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition. Containers may explode when heated. Vapors may form explosive mixtures with air.

### Hazardous Combustion Products

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Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Gaseous hydrogen fluoride (HF).

## **5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **6.1. Personal precautions, protective equipment and emergency procedures**

Remove all sources of ignition. Take precautionary measures against static discharges.

### **6.2. Environmental precautions**

See Section 12 for additional Ecological Information.

### **6.3. Methods and material for containment and cleaning up**

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### **6.4. Reference to other sections**

Refer to protective measures listed in Sections 8 and 13.

## **SECTION 7: HANDLING AND STORAGE**

### **7.1. Precautions for safe handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin and eyes. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Use spark-proof tools and explosion-proof equipment. Use only non-sparking tools. Keep away from open flames, hot surfaces and sources of ignition. Take precautionary measures against static discharges.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### **7.2. Conditions for safe storage, including any incompatibilities**

Keep in a dry place. Keep container tightly closed. Keep away from heat, sparks and flame. Refrigerator/flammables. Keep container tightly closed in a dry and well-ventilated place.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Class 3

**Switzerland - Storage of hazardous substances**

Storage class - SC 3  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### **7.3. Specific end use(s)**

Use in laboratories

## **SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

### **8.1. Control parameters**

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## Exposure limits

List source(s):

| Component                                  | Russia                                    | Slovak Republic | Slovenia | Sweden | Turkey |
|--------------------------------------------|-------------------------------------------|-----------------|----------|--------|--------|
| 1,2-Benzenediol,<br>4-(1,1-dimethylethyl)- | Skin notation<br>MAC: 2 mg/m <sup>3</sup> |                 |          |        |        |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                                                     | Acute effects local<br>(Inhalation) | Acute effects<br>systemic (Inhalation) | Chronic effects local<br>(Inhalation) | Chronic effects<br>systemic (Inhalation) |
|---------------------------------------------------------------|-------------------------------------|----------------------------------------|---------------------------------------|------------------------------------------|
| 1,2-Benzenediol,<br>4-(1,1-dimethylethyl)-<br>98-29-3 ( 0.1 ) |                                     |                                        |                                       | DNEL = 1.6mg/m <sup>3</sup>              |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                                                     | Fresh water    | Fresh water<br>sediment        | Water Intermittent | Microorganisms in<br>sewage treatment | Soil (Agriculture)          |
|---------------------------------------------------------------|----------------|--------------------------------|--------------------|---------------------------------------|-----------------------------|
| 1,2-Benzenediol,<br>4-(1,1-dimethylethyl)-<br>98-29-3 ( 0.1 ) | PNEC = 1.2µg/L | PNEC = 6.9µg/kg<br>sediment dw | PNEC = 1.2µg/L     | PNEC = 0.16mg/L                       | PNEC = 0.68µg/kg<br>soil dw |

| Component                                                     | Marine water    | Marine water<br>sediment        | Marine water<br>Intermittent | Food chain | Air |
|---------------------------------------------------------------|-----------------|---------------------------------|------------------------------|------------|-----|
| 1,2-Benzenediol,<br>4-(1,1-dimethylethyl)-<br>98-29-3 ( 0.1 ) | PNEC = 0.12µg/L | PNEC = 0.69µg/kg<br>sediment dw |                              |            |     |

## 8.2. Exposure controls

### Engineering Measures

Use explosion-proof electrical/ventilating/lighting/equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

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Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

## Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Viton (R)      | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls** No information available.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                               |                                          |
|------------------------------------------------|-------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                        |                                          |
| <b>Appearance</b>                              | Clear                         |                                          |
| <b>Odor</b>                                    | No information available      |                                          |
| <b>Odor Threshold</b>                          | No data available             |                                          |
| <b>Melting Point/Range</b>                     | No data available             |                                          |
| <b>Softening Point</b>                         | No data available             |                                          |
| <b>Boiling Point/Range</b>                     | 145 - 148 °C / 293 - 298.4 °F | @ 760 mmHg                               |
| <b>Flammability (liquid)</b>                   | Flammable                     | On basis of test data                    |
| <b>Flammability (solid,gas)</b>                | Not applicable                | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available             |                                          |
| <b>Flash Point</b>                             | 26 °C / 78.8 °F               | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available             |                                          |
| <b>Decomposition Temperature</b>               | No data available             |                                          |
| <b>pH</b>                                      | No information available      |                                          |
| <b>Viscosity</b>                               | No data available             |                                          |
| <b>Water Solubility</b>                        | No information available      |                                          |
| <b>Solubility in other solvents</b>            | No information available      |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                               |                                          |
| <b>Component</b>                               | <b>log Pow</b>                |                                          |
| 1,2-Benzenediol, 4-(1,1-dimethylethyl)-        | 1.98                          |                                          |

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|                            |                         |             |
|----------------------------|-------------------------|-------------|
| Vapor Pressure             | No data available       |             |
| Density / Specific Gravity | 1.024                   |             |
| Bulk Density               | Not applicable          | Liquid      |
| Vapor Density              | No data available       | (Air = 1.0) |
| Particle characteristics   | Not applicable (liquid) |             |

## 9.2. Other information

|                      |                                        |
|----------------------|----------------------------------------|
| Molecular Formula    | C8 H7 F                                |
| Molecular Weight     | 122.14                                 |
| Explosive Properties | explosive air/vapour mixtures possible |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

|                          |                           |
|--------------------------|---------------------------|
| Hazardous Polymerization | No information available. |
| Hazardous Reactions      | No information available. |

### 10.4. Conditions to avoid

Keep away from open flames, hot surfaces and sources of ignition. Exposure to light.  
Incompatible products.

### 10.5. Incompatible materials

Acids. Bases. Strong oxidizing agents. Halogens. Reducing Agent.

### 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Gaseous hydrogen fluoride (HF).

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

|            |                                                                  |
|------------|------------------------------------------------------------------|
| Oral       | Based on available data, the classification criteria are not met |
| Dermal     | Based on available data, the classification criteria are not met |
| Inhalation | Based on available data, the classification criteria are not met |

| Component                               | LD50 Oral         | LD50 Dermal        | LC50 Inhalation |
|-----------------------------------------|-------------------|--------------------|-----------------|
| 1,2-Benzenediol, 4-(1,1-dimethylethyl)- | 815 mg/kg ( Rat ) | 1331 mg/kg ( Rat ) | -               |

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 2

#### (d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

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|                                            |                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| (e) germ cell mutagenicity;                | No data available                                                                                                    |
| (f) carcinogenicity;                       | No data available<br>There are no known carcinogenic chemicals in this product                                       |
| (g) reproductive toxicity;                 | No data available                                                                                                    |
| (h) STOT-single exposure;                  | Category 3                                                                                                           |
| Results / Target organs                    | Respiratory system.                                                                                                  |
| (i) STOT-repeated exposure;                | No data available                                                                                                    |
| Target Organs                              | No information available.                                                                                            |
| (j) aspiration hazard;                     | No data available                                                                                                    |
| Other Adverse Effects                      | The toxicological properties have not been fully investigated.                                                       |
| Symptoms / effects, both acute and delayed | Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. |

## 11.2. Information on other hazards

|                                                                                               |                                                                                 |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Endocrine Disrupting Properties<br>Assess endocrine disrupting<br>properties for human health | .<br>Contains a substance on the National Authorities Endocrine Disruptor Lists |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

Ecotoxicity effects Do not empty into drains.

| Component                               | Freshwater Fish      | Water Flea         | Freshwater Algae |
|-----------------------------------------|----------------------|--------------------|------------------|
| 1,2-Benzenediol, 4-(1,1-dimethylethyl)- | LC50 = 0.12 mg/L 96h | EC50=0.48 mg/L 48h |                  |

| Component                               | Microtox | M-Factor |
|-----------------------------------------|----------|----------|
| 1,2-Benzenediol, 4-(1,1-dimethylethyl)- |          | 1        |

12.2. Persistence and degradability No information available

12.3. Bioaccumulative potential No information available

| Component                               | log Pow | Bioconcentration factor (BCF) |
|-----------------------------------------|---------|-------------------------------|
| 1,2-Benzenediol, 4-(1,1-dimethylethyl)- | 1.98    | No data available             |

12.4. Mobility in soil No information available

12.5. Results of PBT and vPvB assessment No data available for assessment.

### 12.6. Endocrine disrupting properties

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

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**Assess endocrine disrupting properties for the environment**

Contains a substance on the National Authorities Endocrine Disruptor Lists.

**12.7. Other adverse effects**  
**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

**Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**European Waste Catalogue (EWC)**

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.

**Switzerland - Waste Ordinance**

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

**14.1. UN number**

UN1993

**14.2. UN proper shipping name**

Flammable liquid, n.o.s.  
(4-FLUOROSTYRENE)

**Technical Shipping Name**  
**14.3. Transport hazard class(es)**

3

**14.4. Packing group**

III

### ADR

**14.1. UN number**

UN1993

**14.2. UN proper shipping name**

Flammable liquid, n.o.s.  
(4-FLUOROSTYRENE)

**Technical Shipping Name**  
**14.3. Transport hazard class(es)**

3

**14.4. Packing group**

III

### IATA

**14.1. UN number**

UN1993

**14.2. UN proper shipping name**

Flammable liquid, n.o.s.  
(4-FLUOROSTYRENE)

**Technical Shipping Name**  
**14.3. Transport hazard class(es)**

3

**14.4. Packing group**

III

**14.5. Environmental hazards**

No hazards identified

**14.6. Special precautions for user**

No special precautions required.

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## 14.7. Maritime transport in bulk according to IMO instruments

Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                               | CAS No   | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------------------------------------|----------|-----------|--------|-----|-------|------|----------|------|------|
| 4-Fluorostyrene                         | 405-99-2 | 206-975-0 | -      | -   | -     | X    | -        | -    | -    |
| 1,2-Benzenediol, 4-(1,1-dimethylethyl)- | 98-29-3  | 202-653-9 | -      | -   | X     | X    | KE-11368 | X    | X    |

| Component                               | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------------------------------------|----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| 4-Fluorostyrene                         | 405-99-2 | -    | -                                             | -   | -    | -    | -     | -     |
| 1,2-Benzenediol, 4-(1,1-dimethylethyl)- | 98-29-3  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component                               | CAS No   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------------------------------|----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 4-Fluorostyrene                         | 405-99-2 | -                                                                   | -                                                                             | -                                                                                                     |
| 1,2-Benzenediol, 4-(1,1-dimethylethyl)- | 98-29-3  | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

#### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

### Seveso III Directive (2012/18/EC)

| Component                               | CAS No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------------------------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 4-Fluorostyrene                         | 405-99-2 | Not applicable                                                                            | Not applicable                                                                           |
| 1,2-Benzenediol, 4-(1,1-dimethylethyl)- | 98-29-3  | Not applicable                                                                            | Not applicable                                                                           |

### Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

### Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

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## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

**WGK Classification** See table for values

| Component                                  | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|--------------------------------------------|---------------------------------------|-------------------------|
| 1,2-Benzenediol,<br>4-(1,1-dimethylethyl)- | WGK3                                  |                         |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                                                  | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 1,2-Benzenediol, 4-(1,1-dimethylethyl)-<br>98-29-3 ( 0.1 ) | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H302 - Harmful if swallowed  
H312 - Harmful in contact with skin  
H314 - Causes severe skin burns and eye damage  
H315 - Causes skin irritation  
H317 - May cause an allergic skin reaction  
H318 - Causes serious eye damage  
H319 - Causes serious eye irritation  
H335 - May cause respiratory irritation  
H400 - Very toxic to aquatic life  
H410 - Very toxic to aquatic life with long lasting effects

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

# SAFETY DATA SHEET

4-Fluorostyrene

Revision Date 07-Feb-2024

Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development  
**BCF** - Bioconcentration factor

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

**Prepared By**

Health, Safety and Environmental Department

**Revision Date**

07-Feb-2024

**Revision Summary**

New emergency telephone response service provider.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

**Disclaimer**

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**End of Safety Data Sheet**